

CS257 Linear and Convex Optimization

Homework 5

Due: October 26, 2020

October 19, 2020

1. Convert the following LP problem into standard form.

$$\begin{aligned} \max_{x_1, x_2} \quad & 2x_1 - 3x_2 \\ \text{s.t.} \quad & x_1 + x_2 = 7 \\ & x_1 - 2x_2 \leq 4 \\ & x_1 \geq 0 \end{aligned}$$

Solution. Convert maximization problem into minimization problem.

$$\begin{aligned} \min_{x_1, x_2} \quad & -2x_1 + 3x_2 \\ \text{s.t.} \quad & x_1 + x_2 = 7 \\ & x_1 - 2x_2 \leq 4 \\ & x_1 \geq 0 \end{aligned}$$

Split x_2 into positive and negative parts $x_2 = x_2^+ - x_2^-$.

$$\begin{aligned} \min_{x_1, x_2^+, x_2^-} \quad & -2x_1 + 3x_2^+ - 3x_2^- \\ \text{s.t.} \quad & x_1 + x_2^+ - x_2^- = 7 \\ & x_1 - 2x_2^+ + 2x_2^- \leq 4 \\ & x_1 \geq 0, x_2^+ \geq 0, x_2^- \geq 0 \end{aligned}$$

Introduce slack variable s to convert inequality constraint to equality constraint.

$$\begin{aligned} \min_{x_1, x_2^+, x_2^-, s} \quad & -2x_1 + 3x_2^+ - 3x_2^- \\ \text{s.t.} \quad & x_1 + x_2^+ - x_2^- = 7 \\ & x_1 - 2x_2^+ + 2x_2^- + s = 4 \\ & x_1 \geq 0, x_2^+ \geq 0, x_2^- \geq 0, s \geq 0 \end{aligned}$$

□

2. Consider the optimization problem

$$\begin{aligned} \min_{x_1, x_2} \quad & f(x_1, x_2) \\ \text{s.t.} \quad & 2x_1 + x_2 \geq 1 \\ & x_1 + 3x_2 \geq 1 \\ & x_1 \geq 0, x_2 \geq 0 \end{aligned}$$

Sketch the feasible set. For the objective functions in (a)-(c), first find the set of optimal solutions and the optimal value graphically, and then solve them using CVXPY. For the objective functions in (d) and (e), solve them using CVXPY. Show the outputs of CVXPY.

- (a). $f(x_1, x_2) = x_1 + x_2$
- (b). $f(x_1, x_2) = -x_1 - x_2$
- (c). $f(x_1, x_2) = x_1$
- (d). $f(x_1, x_2) = \max\{x_1, x_2\}$
- (e). $f(x_1, x_2) = x_1^2 + 9x_2^2$

Solution. The feasible set is the shaded region in Figure 1.

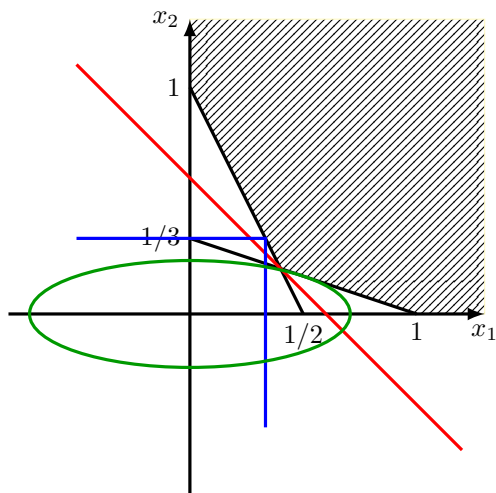


Figure 1: Problem 1

- (a). $x^* = (2/5, 1/5)$, $f^* = 3/5$. See the red line in Figure 1.

The optimal value is 0.6
A solution x is [0.4 0.2]

- (b). No optimal solution. $f^* = -\infty$.

The optimal value is -inf
A solution x is None

(c). $X_{opt} = \{(0, x_2) : x_2 \geq 1\}$, $f^* = 0$.

The optimal value is -9.94411737032918e-05
 A solution x is [-9.94411737e-05 3.40854280e+00]

(d). The optimal value is 0.3333333333333333

A solution x is [0.33333333 0.33333333]

The blue curve in Figure 1 is the level set corresponding to the optimal value.

(e). The optimal value is 0.5

A solution x is [0.5 0.16666667]

The green curve in Figure 1 is the level set corresponding to the optimal value.

□

3. Consider the following optimization problem with $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\mathbf{b} \in \mathbb{R}^m$.

$$\begin{aligned} \min_{\mathbf{x}} \quad & \|\mathbf{Ax} - \mathbf{b}\|_1 \\ \text{s.t.} \quad & \|\mathbf{x}\|_\infty \leq 1 \end{aligned} \tag{1}$$

(a). Reformulate problem (1) as an LP.

(b). Solve the original problem in (1) using CVXPY for $m = 3$, $n = 2$,

$$\mathbf{A} = \begin{pmatrix} 3 & 2 \\ 2 & -4 \\ 2 & 5 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 10 \\ 12 \\ -6 \end{pmatrix}.$$

Show the output of CVXPY.

(c). Solve the LP problem you find in part (a) using CVXPY. Show the output of CVXPY.

Solution. (a). Introduce auxiliary variables $\mathbf{y} \in \mathbb{R}^m$.

$$\begin{aligned} \min_{\mathbf{x}, \mathbf{y}} \quad & \mathbf{1}^T \mathbf{y} \\ \text{s.t.} \quad & -\mathbf{y} \leq \mathbf{Ax} - \mathbf{b} \leq \mathbf{y} \\ & -\mathbf{1} \leq \mathbf{x} \leq \mathbf{1} \end{aligned}$$

(b). The optimal value is 17.999999970151602

A solution x is [1. -1.]

(c). The optimal value is 17.999999999831584

A solution is x = [1. -1.], y= [9. 6. 3.]

□

4. Let $\mathbf{w} \in \mathbb{R}^2$ and

$$\mathbf{X} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} 3 \\ 2 \\ 2 \end{pmatrix}.$$

Solve problem (a) by hand and problems (b), (c) with CVXPY and show the outputs.

(a). Linear least squares regression

$$\min_{\mathbf{w}} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2$$

(b). Lasso

$$\begin{aligned} \min_{\mathbf{w}} \quad & \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 \\ \text{s.t.} \quad & \|\mathbf{w}\|_1 \leq t \end{aligned}$$

for $t = 1$ and $t = 10$. In each case, is the solution the same as that of (a)? Is the solution sparse? Here 'sparse' means that, at a solution $\mathbf{w}^*$, we will have most components of $\mathbf{w}^*$ close to zero. (Note that there are numerical errors, which you should ignore when answering the questions.)

(c). Ridge regression

$$\begin{aligned} \min_{\mathbf{w}} \quad & \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 \\ \text{s.t.} \quad & \|\mathbf{w}\|_2^2 \leq t \end{aligned}$$

for $t = 1$ and $t = 100$. In each case, is the solution the same as that of (a)? Is the solution sparse? (Again ignore the numerical errors.)

Remark. To understand why Lasso produces sparse solutions, draw a contour map of the objective function and the feasible region in the $\mathbf{w}$ -plane; see also Figure 2 of the Lasso paper <http://statweb.stanford.edu/~tibs/lasso/lasso.pdf>

Solution. (a). The normal equation is

$$\mathbf{X}^T \mathbf{X} \mathbf{w} = \mathbf{X}^T \mathbf{y},$$

$$\text{with } \mathbf{X}^T \mathbf{X} = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{X}^T \mathbf{y} = (6, 2)^T.$$

$$\mathbf{w}^* = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} = (1.5, 2)^T.$$

(b). For $t = 1$, the optimal value is 9.0000000000000002
A solution $\mathbf{w}$ is [1.00000000e+00 1.32836223e-22]

For $t = 10$, the optimal value is 4.0000000001897815
A solution $\mathbf{w}$ is [1.50000642 2.00001004]

When $t = 1$, the solution is sparse and different from the solution in (a). When $t = 10$, the solution is the same as that in (a) and not sparse.

(c). For $t = 1$, the optimal value is 7.8574896437354
A solution w is [0.86270615 0.50570554]

For $t = 100$, the optimal value is 3.9999999993483915
A solution w is [1.50000161 2.00001107]

When $t = 1$, the solution is different from the solution in (a). When $t = 100$, it is the same as that in (a). In both cases, the solutions are not sparse.

□